1. $Given\ f(x) = x + 3, g(x) = x^2 + 1$ $and\ h(x) = x.\ Find\ foh$
   A. $x^2 + 1$ B. $x^2 + 4$
   C.* $x + 3$ D. $x^2 + 2$

2. $Given\ f(x) = x + 3, g(x) = x^2 + 1$ $and\ h(x) = x.\ Find\ goh$
   A.* $x^2 + 1$ B. $x^2 + 4$
   C. $x + 1$ D. $x^2 + 2$

3. $Given\ f(x) = x + 3, g(x) = x^2 + 1$ $and\ h(x) = x. Find\ gof$
   A. $x^2 + 6x - 10$ B. $x^2 - 6x - 10$
   C. $x^2 - 6x + 10$ D.* $x^2 + 6x + 10$

4. $Given\ f(x) = x + 3, g(x) = x^2 + 1$ $and\ h(x) = x. Find\ hof$
   A. $x - 3$ B. $x^2 + 1$
   C.* $x + 3$ D. $x^2 - 1$

5. $Given\ f(x) = x + 3, g(x) = x^2 + 1$ $and\ h(x) = x. Find\ hog$
   A. $x - 3$ B.* $x^2 + 1$
   C. $x + 3$ D. $x^2 - 1$

6. $Given\ f(x) = x + 3, g(x) = x^2 + 1$ $and\ h(x) = x.\ Find\ fog$
   A. $2x^3 + 4$ B. $2x^2 + 4$
   C.* $x^2 + 4$ D. $2x^2$

7. $Given\ f(x) = x + 3, g(x) = x^2 + 1$ $and\ h(x) = x.\ Find\ fogoh$
   A. $2x^3 + 4$ B. $2x^2 + 4$
   C.* $x^2 + 4$ D. $2x^2$

8. $Evaluate\ |x - 3| > 1$
   A. $x < 2$ B. $x > 4$
   C. $x > 2$ D.* A & B

9. $Evaluate\ |2x - 5| < 3$
   A. $x > 4$ B.* $1 < x < 4$
   C. $x > 1$ D. A & B

10. $Evaluate\ |2 - 4x| \geq 6$
    A. $x \geq 2$ B. $x \leq -1$
    C. $2 \leq x > -1$ D.* A & B

11. $Express\ cos^2\theta\ in\ terms\ of\ multiole\ angles$
    A. $2[cos2\theta - 1]$ B. $2^{-1}[cos2\theta - 1]$
    C. $2[cos2\theta + 1]$ D.* $2^{-1}[cos2\theta + 1]$

12. $Find\ f(x) = \sqrt{x+4}\ when\ f(0)$
    A. -2 B. 0
    C.* 2 D. $\sqrt{2}$

13. $Find\ f(x) = \sqrt{x+4}\ when\ f(-1)$
    A. 0 B. $i + 1$
    C.* $\sqrt{3}$ D. $\sqrt{4}$

14. $Find\ f(x) = \sqrt{x+4}\ when\ f(-2)$
    A. $\sqrt{3}$ B. 0
    C. i D.* $\sqrt{2}$

15. $Find\ f(x) = \sqrt{x+4}\ when\ f(-3)$
    A. i B. $-i$
    C.* 1^2 D. 0

16. $Find\ f(x) = \sqrt{x+4}\ when\ f(-4)$
    A. $\sqrt{2}$ B.* 0
    C. -2 D. i

17. $Given\ that\ f(x) = \sqrt{x+4}. Compute\ f(x) = 2$
    A. 1 B.* 0
    C. $\frac{1}{\sqrt{2}}$ D. i

18. $Given\ that\ f(x) = \sqrt{x+4}. Compute\ f(x) = 1$
    A. $i + 2$ B. 0
    C. 4 D.* -3

19. $What\ is\ the\ value\ of\ a\ given\ a + ib = \frac{1-i}{2+i}$
    A. $-\frac{1}{5}$ B. $\frac{3}{5}$
    C. $-\frac{3}{5}$ D.* $\frac{1}{5}$

20. Given that $Z_1 = 3 + 2i$ and $Z_2 = 4 - 5i$, evaluate $Z_1 Z_2$
    A. $2 - 7i$ B. $22 - 7i$
    C. $2 + 7i$ D.* $22 + 7i$

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21. *If* $Z_1 = 1 - 3i$ *and* $Z_2 = -2 + 5i$, *determine* $\frac{Z_1 Z_2}{Z_1 + Z_2}$ *in the form* $a + ib$

A. $-\frac{9}{5} + \frac{37}{5}i$ B.* $\frac{9}{5} - \frac{37}{5}i$

C. $-\frac{12}{25} - \frac{13}{25}i$ D. $\frac{9}{25} - \frac{37}{25}i$

22. Given the universal set $M = \{1, 2, 3, 4, 5, 6, 7, 8\}$ where $A = \{1, 3, 5, 7\}, B = \{1, 2, 5, 7\}$ *and* $C = \{3, 6, 7, 8\}$ list the members of $A^c \cap B \cap C^c$

A. $\{1\}$ B.* $\{2\}$

C. $\{3\}$ D. $\{4\}$

23. If $X \cap Y = \emptyset$ then X and Y are said to be

A. finite sets B.* disjoint sets

C. null sets D. infinite sets

24. *Solve* $x^2 + 8x + 16 > 0$

A.* $x > -4$ B. $x < 4$

C. $x \geq 2$ D. $x \geq -4$

25. If set $F = \{x : (2 < x \leq 5) \cup (9 < x < 12)\}$. List the members of set F.

A.* $\{3, 4, 5, 10, 11\}$ B. $\{3, 4, 5, 7, 11\}$

C. $\{2, 3, 4, 5, 10\}$ D. $\{3, 4, 5, 6, 7\}$

26. Determine the modulus, argument of $2 + 3i$

A. $\sqrt{13},\ 33.69°$ B.* $\sqrt{13}, 56.31°$

C. $\sqrt{13}, -33.69°$ D. $\sqrt{13}, -56.31°$

27. Determine the modulus, argument of $2 + 4i$.

A.* $2\sqrt{5},\ 63.43°$ B. $2\sqrt{10},\ 63.43°$

C. $2\sqrt{5},\ 26.57°$ D. $2\sqrt{10},\ 26.57°$

28. Solve the inequality $2x - 3 < x + 7$

A. $x + 10 < 0$ B.* $x < 10$

C. $x > -10$ D. $x - 10 > 0$

29. Determine the value of x satisfying $|x + 3| \geq 7$

A. $x \leq -4, x \geq 10$

B.* $x \leq -10, x \geq 4$

C. $x \leq -10, x \geq -4$

D. $x \leq -4, x \geq 10$

30. *Simplify* $\sqrt{20} + \sqrt{125} + \sqrt{45}$

A. $5(\sqrt{5})$ B. $7(\sqrt{5})$

C.* $10(\sqrt{5})$ D. $11(\sqrt{5})$

31. *Simpilfy* $3\sqrt{2} - \sqrt{32} + \sqrt{50}$

A. $5\sqrt{2}$ B. $6\sqrt{2}$

C. $10\sqrt{2}$ D.* $4\sqrt{2}$

32. *Find* $\frac{Z_1}{Z_2}$ *if* $Z_1 = 2 - 4i$ *and* $Z_2 = 6 + 7i$

A. $\left(\frac{16}{85} + i\frac{38}{85}\right)$ B. $\left(\frac{15}{85} - i\frac{38}{85}\right)$

C. $-\left(\frac{16}{85} - i\frac{38}{85}\right)$ D.* $-\left(\frac{16}{85} + i\frac{38}{85}\right)$

33. *Solve* $4x + \frac{1}{3} \geq \frac{2}{3}x + 2$

A. $x \geq 1$ B.* $2x \geq 1$

C. $2x \leq 1$ D. $x \leq 1$

34. Given that $Z_1 = 2 + 3j$ and $Z_2 = 3 - 4j$, evaluate $Z_2 - Z_1$

A. $1 + 7j$ B.* $1 - 7j$

C. $-1 + 7j$ D. $-1 - 7j$

35. Given $Z_1 = 1 + 2i,\ Z_2 = 4 - 3i,\ Z_3 = -2 + 3i$, evaluate $|Z_1 + Z_2| - |Z_3| \times \sqrt{2}$

A. $\sqrt{10}$ B. 10

C. $2\sqrt{5}$ D.* 0

36. If $Z_1 = 1 + 2i$ *and* $Z_2 = 4 - 3i$, what is the value of $|Z_2 - Z_1|$?

A.* $\sqrt{34}$ B. $\sqrt{16}$

C. $2\sqrt{5}$ D. 4

37. Determine the modulus, argument of $-5 - 2i$.

A.* $5.385, -158.2°$ B. $5.385,\ 26.57°$

C. $4.583, -158.2°$ D. $4.583,\ 26.57°$

38. *Given* $w = u + iv, z = x + iy$ *and* $w = \frac{1}{z}$. *find* u.

A.* $\frac{x}{x^2+y^2}$ B. $-\frac{x}{x^2+y^2}$

C. $\frac{y}{x^2+y^2}$ D. $-\frac{y}{x^2+y^2}$

39. *Given* $w = u + iv, z = x + iy$ *and* $w = \frac{1}{z}$, *find* v.
A. $\frac{x}{x^2+y^2}$ B. $-\frac{x}{x^2+y^2}$
C. $\frac{y}{x^2+y^2}$ D.* $-\frac{y}{x^2+y^2}$

40. _____ numbers ranges from $-\infty$ to $+\infty$
A.* real B. integer
C. complex D. natural

41. In a village all the people speak Hausa or English or both. If 97% speak Hausa and 64% speak English, what percentage speak both languages?
A*. 61% B. 63%
C. 67% D. 69%

42. In a village of 650 youths, 250 are girls, 425 of the youths like music. If 90 girls do not like music, how many boys do not like music?
A.* 135 B. 160 C. 175
D. 310 E. 335

43. In a class of 20 boys, 16 play soccer, 12 play hockey and 2 are not allowed to play games. Find how many students play soccer and hockey.
A. 6 B. 8
C.* 10 D. 12

44. Given that $P = \{1,\ 3,\ 5,\ 7\}$ *and* $Q = \{2,\ 5,\ 6,\ 8,\ 9\}$. *Find* $(P \cup Q)$.
A.* {1, 2, 3, 5, 6, 7, 8, 9}
B. {1, 2, 3, 6, 7, 8, 9}
C. {1, 2, 3, 4, 5, 6, 7, 8, 9}
D. {3, 5}

45. The sets $P = \{2, 5, 8\}$ and $Q = \{2, 3, 5, 7\}$ are subsets of the universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. Find $P' \cap Q$
A. {3} B. {3, 5}
C.* {3, 7} D. {1, 3}

46. The sets $P = \{2, 5, 8\}$ and $Q = \{2, 3, 5, 7\}$ are subsets of the universal set $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$.
Find $n(P \cap Q')$
A. 1 B. 2
C. 3 D. 0

47. Given $f(x) = 2x^2$ and $g(x) = 3x + 1$, Compute $(fog)(2)$
A. 25 B. 49
C.* 98 D. 89

48. If $f(x) = 2(x-3)^2 + 3(x-3) + 4$ and $g(y) = 5 + y$. Find $f[g(-1)]$
A. 2 B. 3
C. 4 D.* 9

49. The function of f and g are defined over R as $f: x = x^2 + 2$ and $g: x = \frac{2x}{1-x}$. Find $fog(3)$.
A. -3 B. -9
C. 9 D.* 11

50. If $f: x = 2x - 3$ and $g: x = 3x + p$, for what value of constant p is $fog = gof$.
A. -3 B. 3
C. -2 D.* -6

51. Given $f(x) = 2x^2$ and $g(x) = 3x + 1$, Compute (fog).
A. $9x^2 + 6x + 1$
B. $18x^2 + 12x + 1$
C.* $18x^2 + 12x + 2$
D. $18x^2 + 12x - 2$

52. *Given* $Z = 4 - 2\sqrt{2}i,\ find\ Z\bar{Z}$
A. 21 B. 26
C.* 24 D. 28

53. Determine the modulus, argument of $i(2 - i)$.
A. 1.732, 63.43°
B.* 2.236, 63.43°
C. 1.732, 26.57°
D. 2.236, 26.57°

54. Find the range of values of x for which the inequality $\frac{1}{3}x - 2 \leq \frac{2}{5}x + \frac{1}{4}$

A. $x \geq -\frac{4}{135}$ B.* $x \geq -\frac{135}{4}$

C. $x \geq \frac{4}{135}$ D. $x \geq \frac{135}{4}$

55. *Given the function* $f(x) = 5x - 2$, *find* $f^{-1}(-1)$

A.* $\frac{1}{5}$ B. -7

C. $-\frac{1}{7}$ D. $-\frac{1}{5}$

56. *Find the inverse of* $f: x = \frac{x+1}{x-2}$

A.* $\frac{2x+1}{x-1}$ B. $\frac{2x-1}{x-1}$

C. $\frac{2x+1}{x+1}$ D. $\frac{x-1}{2x+1}$

57. *Find the solution of the inequality* $(12 - 4x) \geq 6$

A. $x \geq \frac{3}{2}$ B. $x \geq -\frac{3}{2}$

C.* $x \leq \frac{3}{2}$ D. $x \leq -\frac{3}{2}$

58. *If* $X = \{2, 4, 6\}$, *determine the number of power set X denoted as* $nP(X)$.

A. 3 B. 6

C.* 8 D. 16

59. *Suppose set* $R = \{2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$ *find the number of element of* R.

A. 2 B. 8

C.* 10 D. 20

60. *If* $z = cos\theta + isin\theta$, *then find the value of* $\left(z + \frac{1}{z}\right)^2$ *is*

A. $2cos\theta$ B. $2isin\theta$

C.* $4cos^2\theta$ D. $-4sin^2\theta$

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